Shivam Kumar

Aspiring AI/ML Engineer | **Generative AI & NLP** | **Applied Machine Learning**

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Projects

RAG-Based AI Teaching Assistant | GitHub — Jan. 2026 – Jul. 2026

- Architected an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline in **Python** using embeddings, **FAISS** vector search, **Hugging Face Transformers**, and **LLMs** for context-aware academic question answering.
- Integrated **Whisper** speech-to-text with semantic retrieval and LLM response generation, **reducing irrelevant responses by 30%** through optimized retrieval logic and prompt engineering.
- Deployed a production-ready **Gradio** application on **Hugging Face Spaces**, enabling real-time text and voice-based Q&A for end users.

Insta Manager Bot – Instagram Automation Tool | GitHub — Apr. 2025 – Dec. 2025

- Engineered a **Flask**-based automation platform for Instagram profile monitoring, engagement analytics, and actionable account insights.
- Automated login and data-collection workflows with **SQLite**-backed persistence, eliminating repetitive manual monitoring and improving data reliability.
- Designed a responsive analytics dashboard visualizing engagement metrics for centralized, real-time account tracking.

IPL Cricket Data Analysis (2008–2024) | GitHub — Feb. 2025 – Mar. 2025

- Conducted **exploratory data analysis (EDA)** on 17 seasons of IPL match data using **Python, Pandas, Matplotlib**, and Jupyter Notebook.
- Uncovered player performance, venue, and match-outcome trends, translating raw datasets into actionable, data-driven insights.
- Built visualizations highlighting top performers and venue trends to support data-driven storytelling and analysis.

Technical Skills

Languages & Core CS: Python, SQL, Data Structures & Algorithms, Object-Oriented Programming (OOP), Git, GitHub

Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Feature Engineering, Model Training & Evaluation, Cross-Validation, Hyperparameter Tuning, Scikit-learn

Deep Learning & NLP: Neural Networks, PyTorch, TensorFlow, Transformers, Transfer Learning, Model Fine-Tuning, NLP, Whisper, Text Embeddings, Semantic Search

Generative AI & LLM Engineering: LLMs, RAG (Retrieval-Augmented Generation), Prompt Engineering, LangChain, FAISS, Vector Databases, LLM APIs, Function Calling, AI Agents, Agentic AI, MCP, Structured Outputs

Data & Backend: NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook, Flask, FastAPI, Hugging Face Transformers, REST APIs, PostgreSQL, MongoDB, SQLite

MLOps & Deployment: Docker, Hugging Face Spaces, AWS, GitHub Actions, CI/CD, Model & Cloud Deployment

Education

Sage University — Jun. 2024 – Jun. 2028
B.Tech. in Artificial Intelligence & Machine Learning | *CGPA: 7.57/10* — *Indore, India*

Certifications

- **Python Programming with Red Hat (AD141-RHA), Ver. 9.0** — Dec. 2025
- **PMR Certification – SAMTRIX** — Dec. 2025
- **AWS Academy Graduate – Machine Learning Foundations** — Dec. 2025

Achievements

- Recognized for strong performance in **AI and coding projects** during the first year of engineering.
- Published and actively maintain multiple **real-world software and AI projects** on **GitHub**, demonstrating hands-on development and end-to-end project ownership.